University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2017, Part 1

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let G be a finite p-group and suppose that G acts on a finite set S. Denote by S^G the subset of S fixed by every element of G. Show that $|S| \equiv |S^G| \pmod{p}$, where $|A|$ denotes the cardinality of the set A.

2. This problem has two parts.

(i) How many elements of S_6 have order 2?

(ii) How many elements of A_6 have order 2?

3. Let G be a finite group, and suppose d is an integer dividing the order of G.

(i) Show the number of elements of order d is divisible by $\phi(d)$ (Euler ϕ-function).

(ii) Give an example in which the number of elements of order d is zero.

(iii) Give an example in which the number of elements of order d is strictly larger than $\phi(d)$.

4. Show that a group of order $385 = 5 \cdot 7 \cdot 11$ has a normal cyclic subgroup of order 77.

5. Let G be a group that is the internal direct product of two characteristic subgroups H and K. Show that $\mathrm{Aut}(G) \simeq \mathrm{Aut}(H) \times \mathrm{Aut}(K)$. Can the hypothesis that H and K are characteristic be omitted?

6. Let p be a prime and $n \geq 3$ an integer. Show that there exist nonabelian groups of order p^n with a cyclic subgroup of index p.